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NIHR Research for Patient Benefit Programme

Scheme	Research for Patient Benefit Programme
Funding Opportunity	2025/425 RfPB Researcher-led

SUMMARY INFORMATION

Application Title	Feasibility and Validation for a AI-enabled Digital Sleep Diary for NHS Sleep Services
Contracting Organisation	University College London Hospitals
Contracting Organisation Department	Sleep Services
Duration (months)	9
Total Cost to NIHR	
Has this application been previously submitted to this or any other funding body?	Yes **Grant Reference**: NIHR511209 **Funder**: 2025/334 NIHR - Research for Patient Benefit Researcher-Led **Please Provide Details**: Incorrect contracting organisation as Lead Applicant. We unfortunately made an error with applying through the incorrect contracting organisation (an SME). This time round we have a partnership with the University College London Hospitals NHS Trust which are the contracting organisation for this application.

LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

Full Name	Dr Venkata McNeillis
Organisation	Sleep and Health Clinic Ltd
Department	ENT Snoring & Sleep Centre
Proposed Role	Lead clinical governance and patient safety oversight; direct PPI strategy across the study lifecycle; ensures NHS integration; and reviews feasibility and validity outcomes against progression criteria to inform adoption decisions.
ORCID	0009-0004-7611-2639
% FTE Commitment	100.00%

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Co-Applicants

Full Name	Mr Terence Hook
Organisation	Sleep and Health Clinic Ltd
Department	Technical Development
Proposed Role	Technical lead responsible for platform stability, DTAC compliance, cyber security assurance, interoperability readiness and real-time technical monitoring during study delivery.
ORCID	0009-0002-0679-3876
% FTE Commitment	100.00%

Full Name	Mr Maxime Sibille
Organisation	Sleep and Health Clinic Ltd
Department	Programme Management
Proposed Role	Leads programme coordination, milestone delivery and reporting; ensures operational alignment between NHS and technical teams; oversees financial governance and risk management.
ORCID	0009-0001-9205-4676
% FTE Commitment	

Full Name	Miss Maria Bozzini
Organisation	UNIVERSITY COLLEGE LONDON HOSPITALS CHARITIES
Department	Sleep Unit
Proposed Role	Leads participant consent, adherence monitoring and actigraphy coordination; manages secure data capture and quality assurance; supports quantitative and qualitative analysis and preparation of outputs to inform progression decisions.
ORCID	0000-0003-2972-3934
% FTE Commitment	30.00%

APPLICATION DETAILS

Plain English Summary

Sleep problems are common and often go undiagnosed. Around one-third adults in the UK report persistent sleep difficulties, and up to 85% of people with obstructive sleep apnoea are thought to remain undiagnosed. Poor sleep is associated with high blood pressure, diabetes, obesity, heart disease, depression and reduced quality of life and difficulty functioning at work. Delays in recognising and assessing sleep disorders can prolong symptoms, increase healthcare use and add pressure to NHS services. In some areas, waits for diagnostic sleep studies now exceed 6 months, and referrals to sleep services have risen sharply in recent years.

In NHS sleep clinical, assessment usually begins with handwritten sleep diaries completed over 14 days. These diaries are often filled in retrospectively, contain missing or unclear entries, and can be difficult for clinicians to interpret. Incomplete or inconsistent data may delay decisions, lead to repeat appointments, and contribute to inefficient use of specialist time.
Smartphone-based digital sleep diaries may offer a more structured and reliable alternative. By enabling near-real-time recording and standardised summaries, they have the potential to support clearer early assessment. However, before a digital tool can be integrated into NHS pathways, it must be shown to be feasible for patients to use, acceptable in routine care, and capable of producing sleep information comparable to established methods.

This study is an early-stage clinical validation within routine NHS care. We will recruit 50 adults aged 18-75 attending the Sleep Unit at University College London Hospitals NHS Foundation Trust. Participants will complete both a digital Sleep Diary and the standard paper diary for 14 days. A subgroup of 20 participants will also wear a wrist-worn sleep monitor to allow comparison with objective sleep measures. We will assess how consistently the digital diary is completed, how usable and acceptable it is, and how closely its results align with paper diaries and wearable data.

Participation is voluntary, and all information will be handled securely in line with NHS and data protection requirements.

We recognise that digital tools must be equitable. Participants will need to own a compatible smartphone and use the app in English. We will record how many people are unable to participate for these reasons, alongside demographic information, to understand potential inequalities and inform future adaptations such as multilingual versions and alternative access options.

People with lived experience of sleep problems have shaped the study design and will continue to contribute throughout. If predefined feasibility, usability and validation criteria are met, the results will inform a larger national study to test whether the digital diary improves NHS referral and triage pathways and represents good value for money.

For patients, a validated digital diary could support clearer early assessment, reduce the need to repeat incomplete diaries, and help ensure people are directed to appropriate care more efficiently. For NHS services, improved early assessment may contribute to better use of specialist clinic capacity and reduced unnecessary follow-up appointments. This study provides the evidence needed to decide whether this approach should be introduced more widely across NHS sleep services.

ACKNOWLEDGEMENT AND CONFLICTS

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I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	The participants declare there are no potential conflicts of interest.

1. Background and Rationale

Sleep disorders are common in the UK and place sustained demand on NHS services. Around one-third of adults report sleep problems, with higher prevalence in women and during perimenopause, contributing to health inequalities. Sleep disturbance is strongly associated with cardiometabolic and mental health disorders, increasing NHS demand.

Diagnostic overlap between insomnia and obstructive sleep apnoea (OSA) contributes to misclassification and delayed treatment. Up to 85% of OSA cases remain undiagnosed, increasing pressure on specialist services.

Current NHS pathways rely primarily on paper-based sleep diaries, most commonly the Consensus Sleep Diary (CSD). Paper diaries are prone to retrospective completion and missing data, reducing triage accuracy. Many commercial sleep apps lack clinical validation, interoperability with NHS systems, and compliance with NHS digital and data security standards, limiting their suitability for routine care.

While electronic sleep diaries have demonstrated improved data completeness in controlled research environments, there is insufficient NHS-based evidence that a digitally governed, interoperable sleep diary can achieve sustained adherence, generate clinically comparable sleep parameters, and integrate safely within NHS governance frameworks. This represents a translational gap between research validation and real-world NHS adoption.

In North Central London, analysis of over 765,000 primary care records reveals a substantial referral gap, with only ~1.7% of patients with recorded insomnia symptoms referred to evidence-based pathways. Regional sleep referrals have increased sharply (>40% year-on-year in some providers) and waits for diagnostic studies frequently exceed 6–12 months. With millions estimated to remain undiagnosed with OSA, latent demand continues to strain NHS sleep services.

1.1 Alignment with the Work and Health Highlight Notice

Sleep disorders are strongly associated with impaired productivity and sickness absence in working-age adults. By improving early classification within NHS pathways, this study supports timely intervention that may reduce prolonged

untreated sleep disturbance and align with national priorities to improve workforce health and reduce economic inactivity.

2. Aims and Objectives

2.1 Overall Aim

To determine whether a Digital Sleep Diary (DSD) can be feasibly integrated within NHS sleep pathways and generate sleep parameters comparable to established measures, thereby informing progression to multi-site diagnostic and health economic evaluation.

2.2 Objectives Table

Objective	*Description*
01	Assess the feasibility via 14-day adherence.
02	Evaluate the usability System Usability Score (SUS) and acceptability.
03	Determine validity versus CSD and actigraphy.
04	Explore user experience and adherence barriers.
05	Identify technical and workflow issues prior to scaling.

Progression to a multi-site implementation and economic evaluation study will only occur if predefined feasibility, usability and agreement thresholds are met.

3. Methodology/ Plan

3.1 Project Design and Methods

This is a non-randomised, single-arm feasibility study designed to assess adherence, usability and preliminary validity.

In line with feasibility study guidance, we aim to recruit 50 participants to allow for attrition. All participants will use the DSD for 14 consecutive days. A subsample of 20 participants will also wear an actigraphy device (CamnTech) during the same period to enable within-participant validation. Although completing both diary formats may increase participant burden, this approach is limited to a short 14-day period and is necessary for direct comparison with established measures.

A sample of 50 allows estimation of adherence rates within 10-15% precision and

provides sufficient observations to calculate agreement parameters Intraclass Correlation Coefficient (ICC) with reasonable stability for progression decision-making. The actigraphy subsample (n=20) is sufficient for exploratory agreement analysis with objective measures while limiting cost and participant burden. This study is not powered for hypothesis testing but is appropriately sized to inform progression decisions.

Baseline demographic data will be collected. Participants will complete both the DSD and paper CSD daily for 14 days; a subsample (n=20) will concurrently wear actigraphy. At study completion, participants will complete the SUS, and a subset will undertake qualitative interviews exploring usability and adherence barriers.

Primary Outcomes:

- *Feasibility*: ≥ 75% mean completion, ≥ 80% retention, and ≤ 10% technical failure.
- *Usability and acceptability*: System Usability Scale (SUS) ≥ 70.
- *Validity:* ICC (two-way mixed, absolute agreement) 0.70 for key sleep parameters, supported by Bland-Altman analysis.

Secondary analyses will explore agreement with actigraphy (n=20), qualitative adherence barriers, and workflow integration issues.

Data will be handled under UCLH governance and GDPR requirements, stored securely on NHS servers with restricted access and encrypted transfer.

3.2 Timeline and Work Packages

M1-2: Ethics approval, DTAC confirmation and study initiation.
M3-5: Recruitment and 14-day collection (digital, paper, actigraphy).
M6-7: Validation (ICC, Bland-Altman) and qualitative analysis.
M8-9: Stakeholder workshop, reporting and progression decision.

3.3 Risk and Mitigation

Key risk includes slower recruitment, suboptimal adherence and technical instability. Recruitment will be supported through parallel clinic streams and weekly screening review. Adherence will be supported through structured onboarding and automated reminders. Platform stability will be monitored in real time with rapid technical response procedures. Predefined progression thresholds ensure that further investment will only proceed if feasibility and agreement criteria are met.

3.4 Study Management & Research Team

Role	*Name*	*Responsibility*
Clinical & PPI Lead	Dr Sara McNeillis	Clinical governance, PPI, NHS integration
Technical Lead	Terry Hook	Platform stability, security, DTAC
Programme Lead	Maxime Sibille	Coordination, milestones, reporting
Research Consultant	Bruna Sanader	Methodology, qualitative supervision
Researcher	Maria Albani	Recruitment, interviews
Research Assistant	Maria Bozzini	Consent, adherence, actigraphy
Statistician	Brian Healy	Agreement modelling (ICC), Bland-Altman analysis, subgroup analysis

The study will operate under UCLH research governance processes with NHS R&D oversight. A Steering Group including clinical, methodological and patient representation will review progress against predefined Go/No-Go criteria.

3.5 Working with People & Communities (PPIE)

Inclusion Criteria

- Adults aged 18-75.
- Able to read and understand English and owns a compatible smartphone (iOS or Android). While smartphone ownership and English fluency are required, findings will inform future adaptations to improve accessibility, including multilingual versions.
- Reports sleep difficulties (e.g., insomnia symptoms) and maintain a consistent sleep-wake schedule (non-shift workers)

Exclusion Criteria

- Current shift workers to avoid confounding effects of circadian disruption and irregular sleep-wake schedules on diary validity.

- Cognitive impairment or untreated psychiatric/ neurological conditions to ensure reliable diary completion and data quality.
- Participation in another interventional sleep study within the past 3 months to minimise carry-over effects and behavioural bias in sleep reporting.

We plan to establish a PPI panel of 8-10 individuals with lived experience, recruited through sleep clinics and patient groups to ensure diversity in age, gender, ethnicity, and socioeconomic background. Three contributors provided initial feedback informing diary simplification and communication procedures.

Participants will be informed of any clinically significant findings. PPI contributors will attend steering meetings, review materials and dissemination outputs, and receive NIHR-recommended reimbursement. Impact will be evaluated via structured feedback.

3.6 Knowledge Mobilisation, Dissemination and Impact

Stakeholders include UCLH clinical leads, ICS digital teams, commissioners and patient contributors. Feasibility findings will be reviewed at midpoint and study completion with UCLH clinical governance and ICS digital leads to inform pathway adoption decisions.

A post-validation workshop will co-develop and Implementation Feasibility Brief outlining workflow integration, governance requirements, interoperability standards and clinician training needs. Barriers and facilitators to adoption will be systematically identified through qualitative interviews and stakeholder engagement to reduce implementation risk.

If predefined feasibility and validity thresholds are met, findings will inform progression to a multi-site diagnostic and formal health economic evaluations (anticipated NIHR Tier 2 application) Preliminary pathway modelling suggests that improved diary completion and earlier classification could reduce repeat consultations and unnecessary specialist referrals, supporting outpatient capacity release and more efficient triage.

Aligned with North Central London ICS priorities on outpatient productivity and digital transformation, the platform will meet NHS DTAC standards to enable pathway adoption.

3.7 Inclusive Research

Recruitment will target diversity across age, sex, gender, ethnicity and socioeconomic status. Sex assigned at birth and gender identity will be collected to enable disaggregated analyses.

Exclusion criteria are evidence-based. Characteristics of excluded individuals will be recorded to monitor bias.

The analysis plan incorporates an equity lens, with subgroup analyses by sex, gender, ethnicity, age, socioeconomic status, and digital access to identify disparities and inform equitable outcomes. Findings will support inclusive clinical recommendations and implementation strategies.

3.8 Intellectual Property (IP) & Commercialisation

The DSD platform and associated software are owned by Sleep and Health Clinic App Limited as background IP. UCLH is acting as the NHS research partner and sponsor site for this validation study under formal collaboration agreements that define roles, responsibilities, data governance, and publication rights.

This RfPB study does not seek to commercialise the platform but to generate independent NHS-based validation evidence required to inform clinical decision-making and progression. All research findings will be published in line with NIHR open access principles.

Should predefined feasibility and validation thresholds be met, future work will explore appropriate regulatory classification, interoperability standards, and NHS adoption pathways. Any future implementation strategy will prioritise patient benefit, NHS affordability, and compliance with digital and data governance standards.

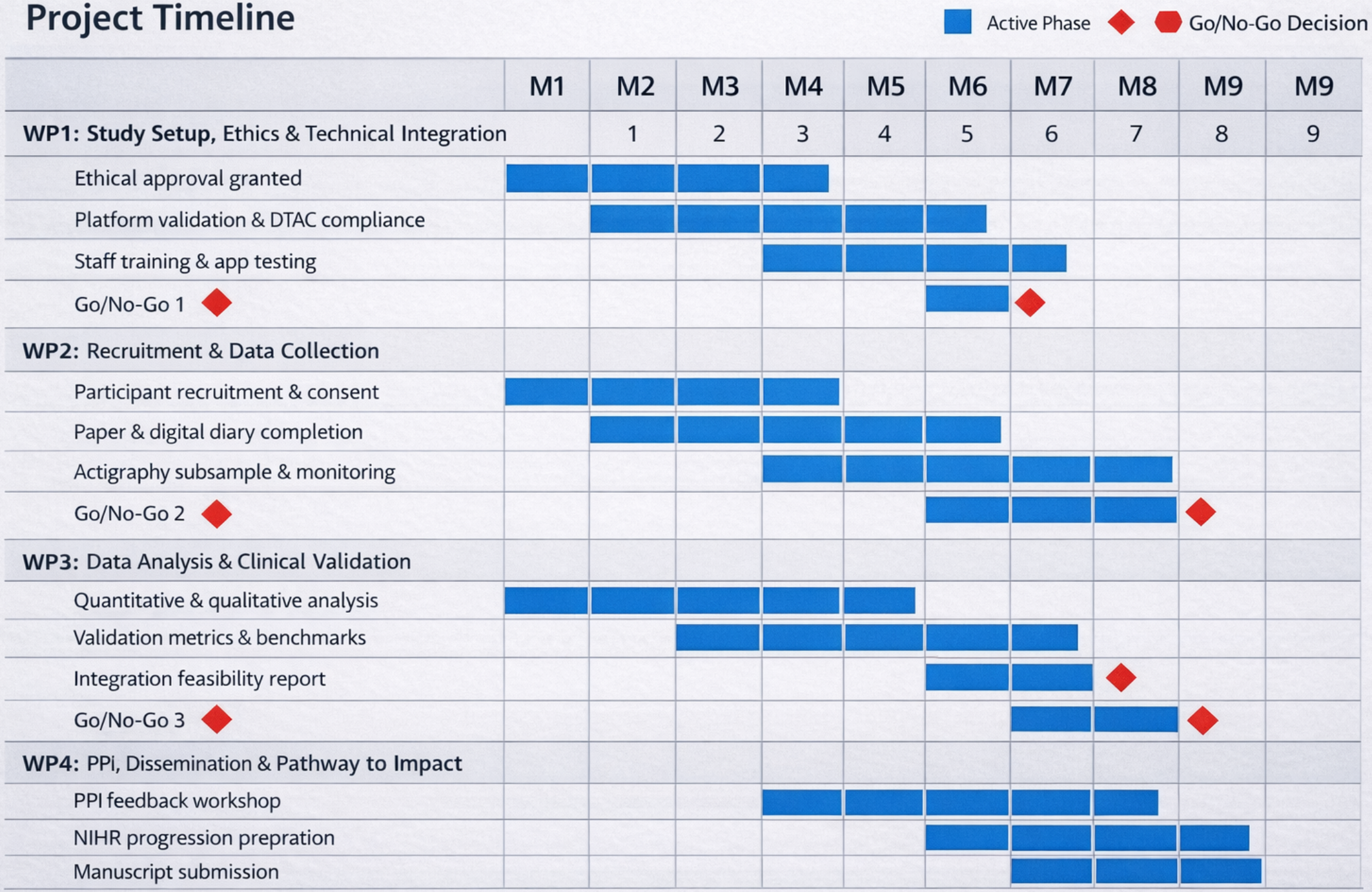
3.9 Value for money

This 9-month study is tightly scoped to determine progression. The sample size (n=50; n=20 actigraphy) is proportionate for feasibility and preliminary validation, balancing methodological rigour with cost efficiency. Actigraphy is limited to a subsample to control costs while enabling objective comparison.

The study used existing NHS infrastructure at UCLH and does not require new capital equipment or service reconfiguration. Defined feasibility and validation thresholds (e.g. adherence 75%, ICC 0.70) provide clear progression criteria, ensuring that further investment will only proceed if justified.

By establishing whether a DSD is usable and produces clinically comparable data, this study de-risks future multi-site and economic evaluation funding, representing a prudent and proportionate use of RfPB resources. In doing so it supports earlier and more accurate assessment of sleep disorders, more efficient use of specialist capacity, and improved health and functioning for patients across the UK.

Project Timeline



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